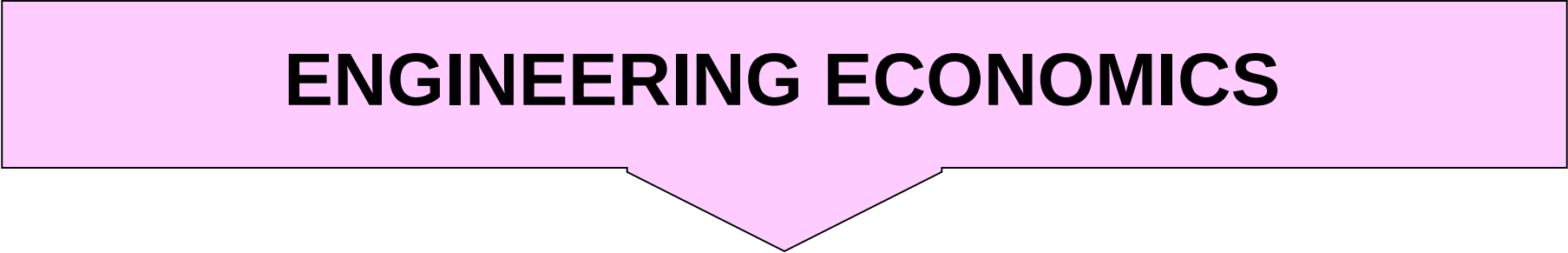
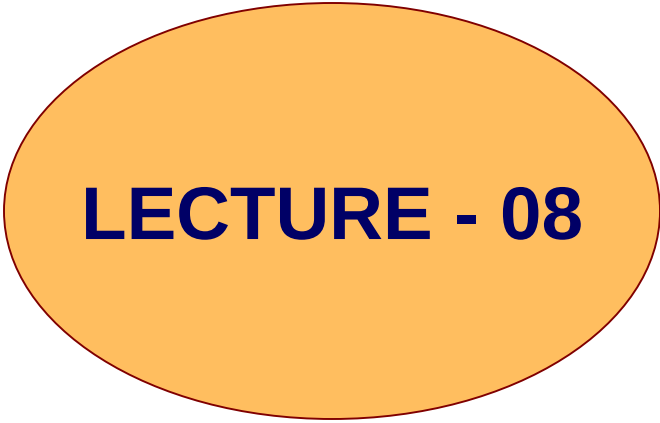


ENGINEERING ECONOMICS



LECTURE - 08



Nominal and Effective Interest Rates

Many financial transactions require that interest be compounded more often than once a year (semi-annually, quarterly, monthly, daily etc). In such situations, there are two expressions for the interest rate.

1. The nominal interest rate, r ,
2. The effective interest rate, i ,

Nominal Interest Rates

The nominal interest rate, r , is expressed on an annual basis: this is the rate that is normally quoted when describing an interest bearing transaction.

Effective Interest Rates

The effective interest rate, i , is the rate that corresponds to the actual interest period. The effective interest rate is obtained by dividing the nominal interest rate by m , the number of interest periods per year.

$$i = r / m$$

Nominal Interest Rates

- The term “nominal” means “in name only”
- In other words, it is **not the real interest rate!**
- We need a way to convert a ***nominal*** interest rate to the true ***effective interest rate*** that will actually apply!

Nominal Interest Rates

$r = (\text{effective interest rate/period}) (\# \text{ of periods})$

- So the *effective interest rate* can be computed as:

$\text{effective interest rate/period} = r/(\# \text{ of periods})$

Effective Interest Rates

- **An effective interest rate is a true, periodic interest rate:**
 - **That applies for a stated period of time**
- **It is conventional to use a year as the standard period of time:**
 - **So, we would like to be able to convert a nominal interest rate to an effective **annual** interest rate**

Examples

- **1.5% per month effective interest rate:**
 - Is the same as $(1.5\%) (12) = 18\%$ nominal interest rate **per year**
- **1% per week effective interest rate:**
 - Is the same as $(1\%) (52) = 52\%$ nominal interest rate **per year**

Problem:

A bank claims to pay interest to its depositors at the rate of 6% per year compounded quarterly. What are the nominal and effective interest rates?

Solution:

a) The nominal interest rate is $r=6\%$

b) Since there are four interest periods per year the effective interest rate is

$$i = r / m$$

$$i = 6\% / 4 = 1.5\% \text{ per quarter}$$

Comparison of Nominal and Effective Interest Rates

It has become customary to quote interest rates on an annual basis, followed by a compounding period if different from one year in length

Comparison of **Nominal and Effective Interest Rates**

Example: If the interest rate is 6% per interest period and the interest period is six months, it is customary to speak of this rate as “12% compounded semiannually.”

Here the annual rate of interest is known as the nominal rate, 12 % in this case. But the actual annual rate on the principle is not 12% but some thing greater, because compounding occur twice during the year.

Consequently, the frequency at which nominal interest rate is compounded each year can have a pronounced effect on the dollar amount of total interest earned

For instance, consider a principal amount of \$1000 to be invested for three years at 12% compounded semiannually

The interest earned during the first six months would be $\$1000 * (0.12/2) = \60 .

Total principal and interest at the beginning of the second six-month period is $P + Pi = \$1000 + \$60 = \$1060$

The interest earned during the second six months would be

$$\textbf{\$1060 * (0.12/2) = \$63.60}$$

The total interest earned during the year is

$$\textbf{\$60.00 + 63.60 = \$123.60}$$

Finally, the effective annual interest rate for the entire year is

$$\textbf{(\$123.60 / \$1000) * 100 = 12.36\%}$$

Discussion